COMPUTING WITH TILES

a wall of Wang tiles that sieves for primes, in ordinary 2″ × 8″ subway tile

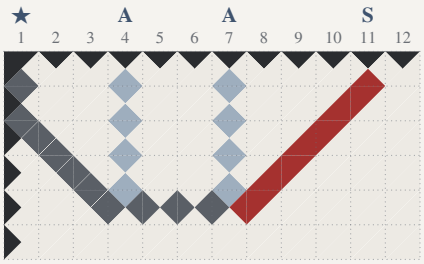
I. WANG TILES

A *Wang tile* is a square with a type on each edge, shown here by color. Tiles are laid edge to edge so that touching edges match, and may not be rotated or reflected. Hao Wang (王浩)† conjectured in 1961 that any set of such tiles that can cover the plane can also cover it periodically. Robert Berger (1966) showed otherwise: tiles can mimic any computer program, one row per step, so whether a set tiles the plane is undecidable. Along the way he found the first *aperiodic* set, 20,426 tiles that tile the plane but never periodically. At right is the smallest aperiodic set possible: 11 tiles with 4 edge types (Jeandel & Rao, 2015).



an aperiodic set: these tile the plane, never periodically

II. TILES THAT ADD

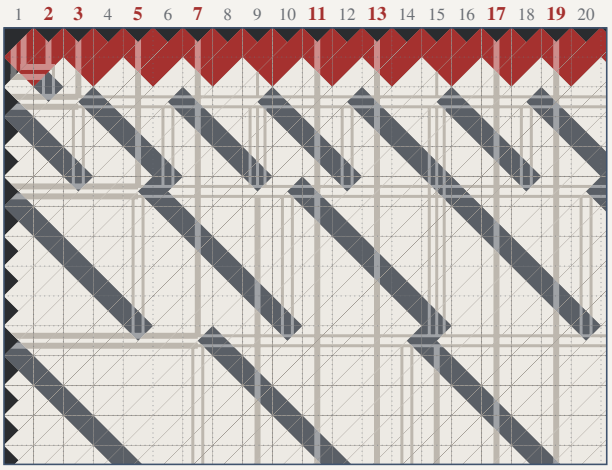


A at 4 and 7 force S to 11

Tiles can do arithmetic because a forced tiling carries information. Both tilings on this page fill a quarter plane whose top and left edges are black: only ★ fits in the corner, and the rest follows. In these 16 tiles (after Grünbaum & Shephard) the edge types draw lines. A light grey line runs down from each **A**; a charcoal diagonal leaves ★ at 45°, turns along the bottom at the first line, and climbs back (red) from the second. Equal diagonals carry the first distance across, so the climb reaches the top at column $A_1 + A_2$: put the **S** anywhere else and the tiling cannot be completed.



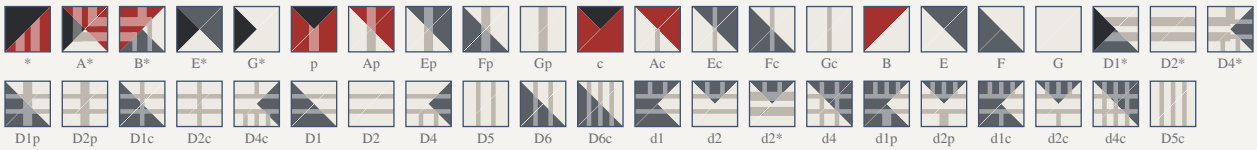
III. A WALL THAT FINDS PRIMES



The wall is a tiling with a set of 43 Wang tiles (section IV). Each edge type is shown by a tile color plus a pattern of etched lines, and a run of matching edges is laid as a single subway tile set on the diagonal. Column *n* stands for the number *n*, and again the black top and left edges force everything else.

The double lines force rows of square boxes with odd sides, repeating across the wall and growing going down: 3, 5, 7, 9, …, each crossed by a charcoal diagonal. The first box in each row, drawn with thick double lines, sends a thick single ray up toward the top. If the ray reaches the top without hitting the edge of a box above it, it marks a prime; if it hits one, it turns into a thin line, marking a composite. (Even numbers get no ray; 2 has a tile of its own.) The thick rays reach the red top row at 2, 3, 5, 7, 11, 13, 17 and 19.

IV. THE 43 TILES



Tile color plus etching gives the edge type, 15 in all; edges must match, and that is the only rule. With the top and left edges black, these tiles fill the quarter plane in exactly one way, and that tiling is aperiodic: its boxes keep growing and the primes never repeat.

† In Cantonese, Wang's surname 王 is a homophone of 黃, Wong: a name held by several members of this house.

Tile set PRIME2b from github.com/cscott/tile-hack (wangprime.js). Addition after Grünbaum & Shephard, Tilings and Patterns, §11.4.

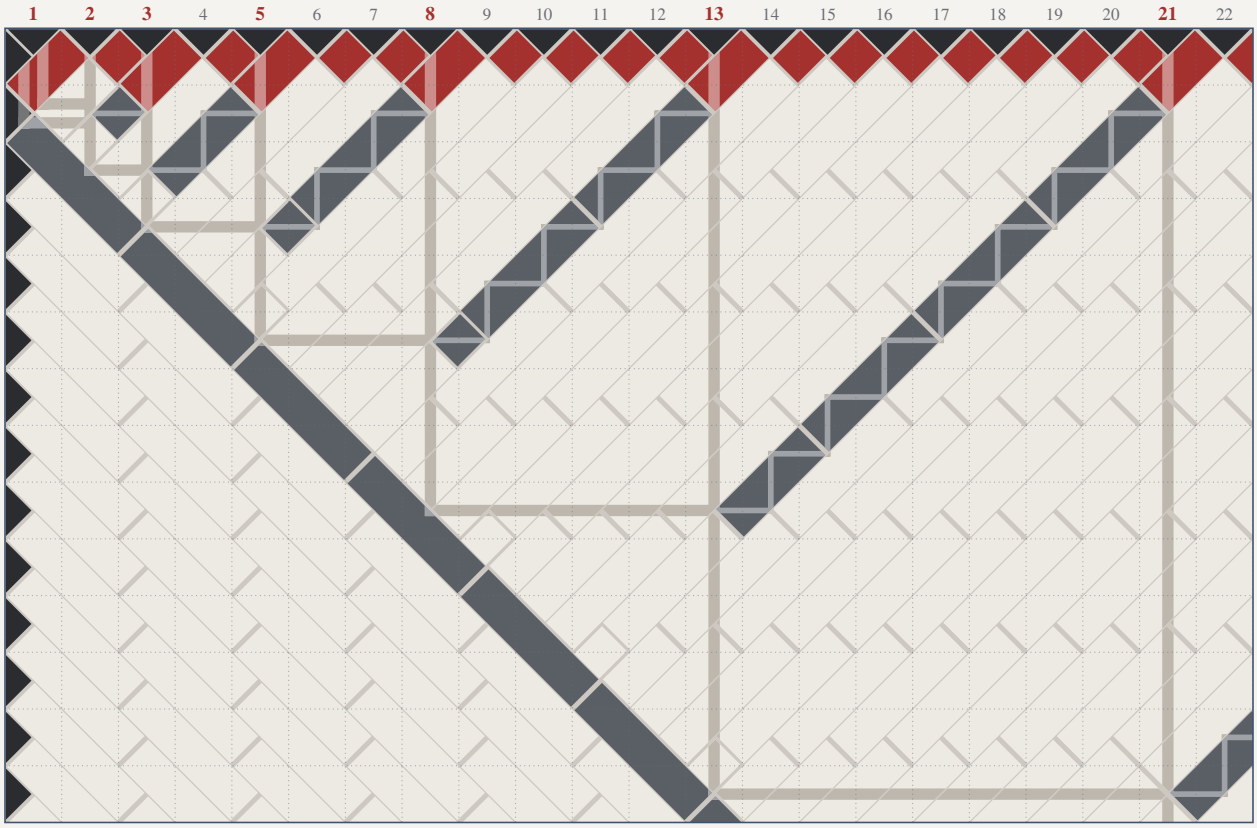
C. Scott Ananian · February 2019

FIBONACCI IN TILE

a second wall of Wang tiles, not yet built: each number the sum of the two before

V. ADDING, AGAIN AND AGAIN

Run the adding machine of section II over and over, and the tiles generate the Fibonacci numbers, each the sum of the two before.[†] Again the top and left edges are black and the tiling is forced; **F′** and **F″** in the corner mark the first two numbers, 1 and 2. Every **F** sends a thick ray straight down. A long charcoal diagonal leaves the corner at 45°, and where it meets the ray of the **F** at column *n*, at depth *n*, the ray turns and runs right to the ray of the next **F**, at column *m*. There a charcoal staircase with a thin etched step climbs back at 45°, rising *n* rows as it moves *n* columns right, and reaches the top at column *m* + *n*: the next **F**.



as it could be laid in 2″ × 8″ subway tile: down-right on and below the long diagonal, up-right above it

VI. THE 18 TILES



Thirteen edge types; edges must match. With the top and left edges black, these tiles fill the quarter plane in exactly one way, and since the Fibonacci numbers spread ever further apart, that tiling is aperiodic. **F′** and **F″** fix the start: add two tiles that let **F″** stand at column 3 instead, and the same machinery produces the Lucas numbers 1, 3, 4, 7, 11, 18, 29.

[†] After Grünbaum & Shephard, *Tilings and Patterns*, Fig. 11.4.2, which has three misprints: its list of tiles leaves out the one that continues the diagonal, its example drops two of the “6” edge labels in column 3, and its caption counts 11 among the Fibonacci numbers.

Tiles after Grünbaum & Shephard, corrected; the layout in tile is greedy, with whole tiles wherever they fit.

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